CHI SQUARE TEST

→ The chi square Test for goodness of fit test claims about population proportions.

→ It is a non parametric test that is performed on categorical [ordinal & nominal] data.

→ There is a population of Male who likes different color bikes.

	Theory	Sample
Yellow Bike	1/3	22
Red Bike	1/3	17
Orange Bike	1/3	59
	⇓	⇓
	Theory Categorical distribution	Observed Categorical distribution

(*) Goodness of fit test

Goodness of fit test:

In Science class of 75 student, 11 are left handed. Does this class fit the theory that 12% of people are left handed.

Ans:

$$\frac{\cancel{12}^{3}}{\cancel{100}_{\cancel{25}_{1}}} \times \cancel{75}^{3} = 9$$

	O (Observed)	E (Expected)
left handed →	11	9
Right handed →	64	66
	75	75

Ex

↓↓
↓↓
↓↓
↓↓
↓↓

→ CHI SQUARE for Goodness Of Fit:

Q. In 2010 the Wiersburg of the city, the weight of the individuals in a small city were found to be the following...

<50 kg	50 - 75	> 75
20 %	30 %	50 %

In 2020, weight of $n = 500$ individuals were sampled. Below are the result

<50	50 - 75	> 75
140	160	200

using $\alpha = 0.05$, would you conclude the population difference of weight has changed in last 10 years?

Ans. :

<50	50 - 75	> 75
20%	30 %.	50%.

2010 - Expected

<50	50-75	> 75
140	160	200

2020 - Observed with $n = 500$

Converted on the basis of 500 sample size

Expected.

<50	50 - 75	> 75
$0.2 \times 500 = 100$	$0.3 \times 500 = 150$	$0.5 \times 500 = 250$

① → Null Hypothesis : H_0 : The data meets the expectation.

→ Alter. Hypothesis : H_1 : The data does not meet the expectation.

② $\alpha = 0.05$ C.I. = 95%

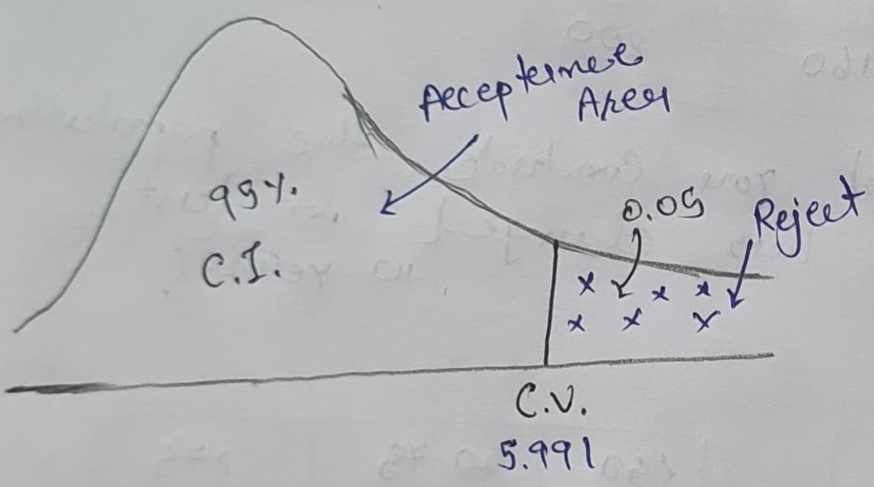
③ Degree of Freedom

$$df = \underset{\text{Category}}{k} - 1 = 3 - 1 = 2$$

④ Decision Boundary

↳ In this we have to find Critical value (c.v.) & if Test statistics is greater than the c.v. then we reject.

→ Here Compare to Z & T test the graph is not like Normal dist.



→ Here.
C.V. = 5.991
From CHI square test table.

→ Assume: If χ^2 is greater than 5.99, Reject Ho

else
we fail to reject Ho

⑤ Calculate CHI square Test statistics.

$$\chi^2 = \sum \frac{(\overset{\text{observed}}{O} - \overset{\text{Expected}}{E})^2}{E}$$

$$= \frac{(140-100)^2}{100} + \frac{(160-150)^2}{150} + \frac{(200-250)^2}{250}$$

$$= \frac{1600}{100} + \frac{100}{150} + \frac{2500}{250}$$

$$= 16 + 0.66 + 10$$

$$= 26.66$$

$$\chi^2 = 26.66$$

$\Rightarrow$ So according to Assumption!

$26.66 > 5.99$